

Gauge Invariance

- Some theories have a redundancy.

In the potential formulation of
Maxwell's fields,

$$\vec{E} = -\vec{\nabla}\phi - \frac{2\vec{A}}{2t}$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

Compactly: $A_\mu = (\phi, -\vec{A})$

$$A_0 = \phi \quad A_i = -\vec{A} \quad i=1,2,3$$

The fields and the phase space
are invariant under

$$A_\mu \rightarrow A_\mu + 2n\lambda$$

$$\phi \rightarrow \phi + 2\pi \lambda$$

" $\frac{2}{2\pi} \lambda$

$$\vec{A} \rightarrow \vec{A} + \vec{\nabla} \lambda$$

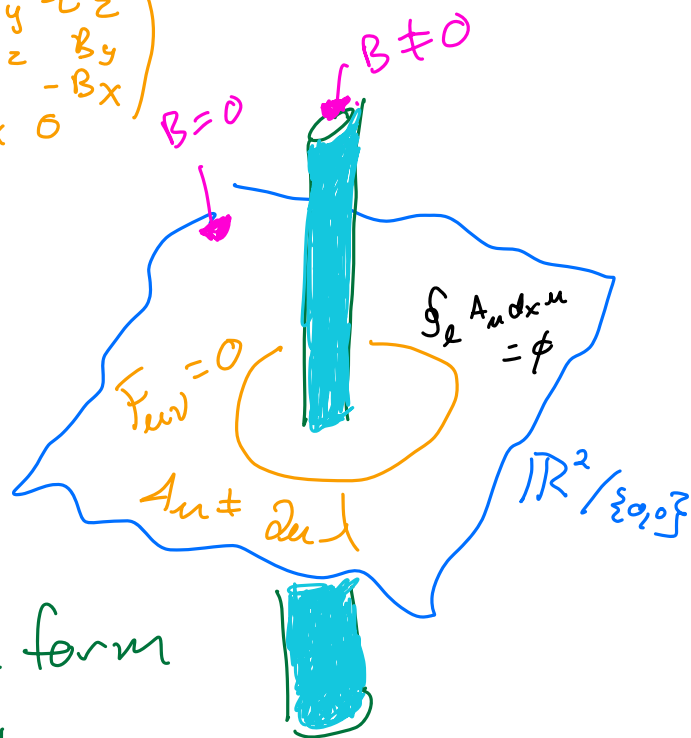
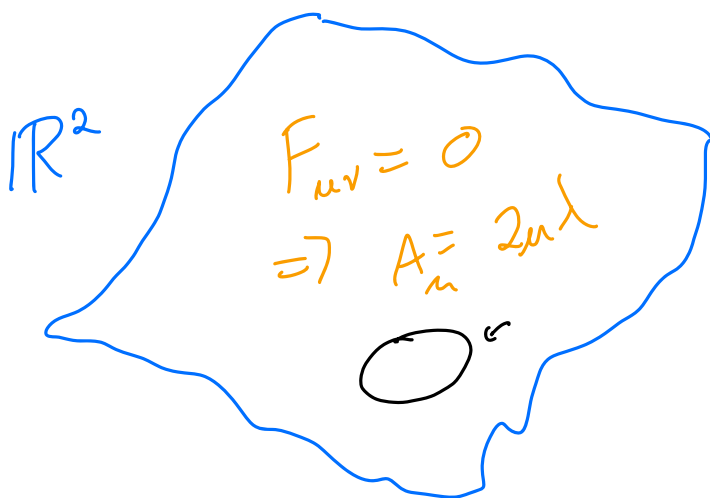
What is a form?

0: $f: M \rightarrow \mathbb{R}$ $\downarrow$ d deth

1: $df = \frac{\partial f}{\partial x^\mu} dx^\mu$ $\downarrow$ Curl

2: $d^2 f = \frac{\partial}{\partial x^\nu} \frac{\partial f}{\partial x^\mu} - \frac{\partial}{\partial x^\mu} \frac{\partial f}{\partial x^\nu} = 0$

flat space $F_{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{pmatrix}$



1 form

$\downarrow$

$$A = A_\mu dx^\mu$$

$$dA = d(A_\mu dx^\mu)$$

$$= \partial_\nu A_\mu dx^\nu \wedge dx^\mu + \partial_\mu A_\nu \underbrace{dx^\mu \wedge dx^\nu}_{-dx^\nu \wedge dx^\mu}$$

$$= (\partial_\nu A_\mu - \partial_\mu A_\nu) dx^\nu \wedge dx^\mu$$

Exterior
Derivative

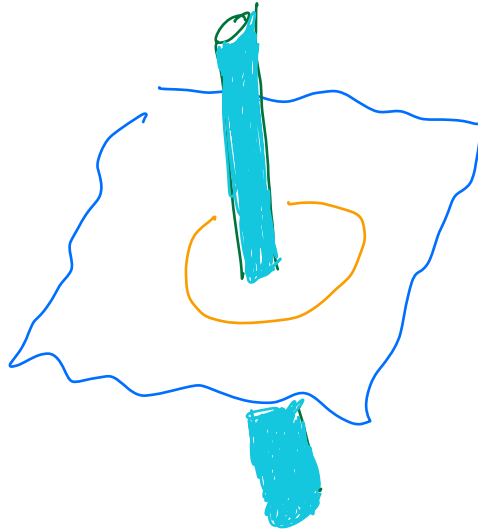
$\nwarrow$

$$\bar{F} = dA$$

In this setting $A = A(x, y) = A_\mu(x, y) dx^\mu$
 $= A_x(x, y) dx + A_y(x, y) dy$

$$A = \frac{\Phi}{2\pi} \frac{x dy - y dx}{x^2 + y^2}$$

$F = dA = 0$
 $A = d\chi$
 $\oint d\chi = 0$

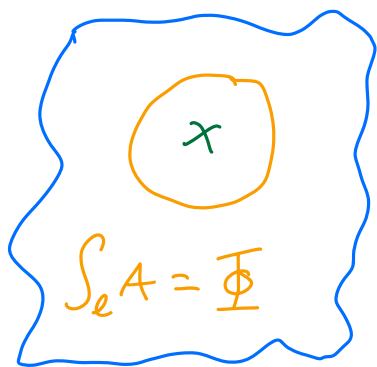


Q: How many gauge-inequivalent $F=0$ vacuum fields are there in EM Maxwell theory?

In other words, how many different A_μ can a theory have s.t. $\tilde{A}_\mu \neq \bar{A}_\mu + d\chi$

$F = dA = 0$
 $A = d\chi$
 $\oint d\chi = 0$

Without defect it is only 1
 since $A = d\chi_1 + d\chi_2$
 $= d(\chi_1 + \chi_2)$



$$A = \frac{\Phi}{2\pi} \frac{x dy - y dx}{x^2 + y^2}$$

$$\int_{\ell} A = \Phi$$

$$\text{since } \int_{\ell} A + d\lambda = \int_{\ell} A_{\ell} + \int d\lambda$$

$$= \int_{\ell} A_{\ell} + \lambda(a) - \lambda(a) = \int_{\ell} A_{\ell}$$

Cohomology definitions

A is a Closed form if

$$dA = 0$$

A is an Exact form if

$$A = d\lambda$$

C_p = space of closed p -forms (vector space)
 E_p = space of exact p -form (vector space)

gauge - inequivalent: $H'_{dR}(M, \mathbb{R}) = C_1 / E_1$

Maybe test?

$$H^0(S^2)$$

$$H^1(S^2)$$

$$H^2(S^2)$$

$$H^0(T^2)$$

$$H^1(T^2)$$

$$H^2(T^2)$$

$\Omega_p(M)$ = space of p forms on M

$$\Omega_{p-1} \xrightarrow{d_p} \Omega_p \xrightarrow{d_{p+1}} \Omega_{p+1} \xrightarrow{d_{p+2}} \Omega_{p+2}$$

$$H_p(M) = \frac{C_p}{E_p} = \frac{\text{Ker}(d_{p+1})}{\text{Im}(d_p)} \quad \leftarrow \text{closed}$$

Cochain Complex Sequence of Abelian groups
& group homomorphisms

$$d^{u+1}: A^u \rightarrow A^{u+1}$$

$$d^{u+1} \circ d^u = 0$$

G is a symmetry of a classical system

In quantum g is represented by a unitary op s.t.

$$[H, U(g)] = 0$$

$$\text{and } \langle \psi | \psi \rangle = \langle \psi | U^\dagger U | \psi \rangle \\ = \langle \psi | \psi \rangle$$

A linear representation is

$$g_1 \circ g_2 = g_3$$

$$U(g_1) \circ U(g_2) = U(g_3) = U(g_1 \circ g_2)$$

However quantum measurements are invariant
up to a phase so quantum states are given
by rays $|\psi\rangle \rightarrow e^{i\alpha} |\psi\rangle$

This allows:

$$U(g_1) U(g_2) |\psi\rangle = U(g_1 \circ g_2) e^{i\phi(g_1, g_2)} |\psi\rangle$$

Can we remove the phase?

$$\begin{array}{c} \Updownarrow \\ [0] \in H^0(G, U(1)) \end{array}$$

$$U(g_1) \circ U(g_2) = e^{i\phi(g_1, g_2)}$$

Try to redefine phase:

$$U'(g) \rightarrow e^{i\alpha(g)} U(g)$$

$$\begin{aligned} U(g_1) e^{i\alpha(g_1)} U(g_2) e^{-i\alpha(g_2)} \\ = e^{i\phi(g_1, g_2) - i\alpha(g_1 \circ g_2)} U(g_1 \circ g_2) \end{aligned}$$

So we need:

$$\alpha(g_1) + \alpha(g_2) - \alpha(g_1 \circ g_2) = -\phi(g_1, g_2)$$

But also since group associativity:

$$U(g_1) (U(g_2) \circ U(g_3)) = (U(g_1) \circ U(g_2)) \cdot U(g_3)$$

$$\begin{aligned} U(g_1) e^{i\phi(g_2, g_3)} U(g_2 \circ g_3) &= e^{i\phi(g_1, g_2)} U(g_1 \circ g_2) U(g_3) \\ e^{i\phi(g_2, g_3)} e^{i\phi(g_1, g_2 \circ g_3)} U(g_1 \circ g_2 \circ g_3) &= e^{i\phi(g_1, g_2) + i\phi(g_1 \circ g_2, g_3)} U(g_1 \circ g_2 \circ g_3) \end{aligned}$$

We also require:

$$0 = \phi(g_2, g_3) + \phi(g_1, g_2 \circ g_3) - \phi(g_1, g_2) - \phi(g_1 \circ g_2, g_3)$$

Group Cohomology can give us inequivalent phases that satisfy:

$$\alpha(g_1) + \alpha(g_2) - \alpha(g_1 \circ g_2) = -\phi(g_1, g_2)$$

and

$$0 = \phi(g_1, g_2) + \phi(g_1 \circ g_2, g_3) - \phi(g_2, g_3) - \phi(g_1, g_2 \circ g_3)$$

$$\alpha: G \rightarrow U(1) \quad \phi: G \times G \rightarrow U(1)$$

Let $C^n(G, A)$ be the group $\underbrace{G \times G \times \dots \times G}_{n\text{-times}} \rightarrow A$

$$\alpha \in C^1(G, U(1)) \quad \phi \in C^2(G, U(1))$$

$$\text{Define } d^{(n+1)}: C^n(G, A) \rightarrow C^{n+1}(G, A)$$

$$\begin{aligned}
 (d^{(n+1)}\psi)(g_1, \dots, g_{n+1}) &= \psi(g_2, \dots, g_{n+1}) \\
 &+ \sum_{i=1}^n (-1)^i \psi(g_1, \dots, g_{i-1}, g_i g_{i+1}, \dots, g_{n+1}) \\
 &+ (-1)^{n+1} \psi(g_1, \dots, g_n)
 \end{aligned}$$

With this we see:

$$(d^2\psi)(g_1, g_2) = \psi(g_2) - \psi(g_1 g_2) + \psi(g_1)$$

$$\alpha(g_1) + \alpha(g_2) - \alpha(g_1 g_2) = -\phi(g_1, g_2)$$

$$\begin{aligned}
 (d^3\psi)(g_1, g_2, g_3) &= \psi(g_2, g_3) - \psi(g_1, g_2, g_3) \\
 &+ \psi(g_1, g_2 g_3) - \psi(g_1, g_2)
 \end{aligned}$$

$$0 = \phi(g_2, g_3) + \phi(g_1, g_2 g_3) - \phi(g_1, g_2) - \phi(g_1 g_2, g_3)$$

To remove a phase $\Phi = -d^{(2)}\alpha$

$$d^{(3)}\Phi = 0$$

We have $d^3 \circ d^2 = 0 \leftarrow$ by

$$(d^2 \psi)(g_1, g_2) = \psi(g_2) - \psi(g_1, g_2) + \psi(g_1)$$

$$(d^3 \psi)(g_1, g_2, g_3) = \psi(g_2, g_3) - \psi(g_1, g_2, g_3) \\ + \psi(g_1, g_2, g_3) - \psi(g_1, g_2)$$

$$\begin{aligned} & \cancel{\psi(g_2)} - \cancel{\psi(g_2 \circ g_3)} + \cancel{\psi(g_3)} - \cancel{\psi(g_1 \circ g_2)} \\ & \quad - \cancel{\psi(g_3)} + \cancel{\psi(g_1 \circ g_2 \circ g_3)} \\ & + \cancel{\psi(g_2 \circ g_3)} - \cancel{\psi(g_1 \circ g_2 \circ g_3)} + \cancel{\psi(g_1)} \\ & \quad - \cancel{\psi(g_2)} + \cancel{\psi(g_1 \circ g_2)} - \cancel{\psi(g_1)} \end{aligned}$$

$$\text{There fore } \phi \in H^2(G, U(1)) = \frac{\text{Ker}(d^3)}{\text{Im}(d^2)}$$

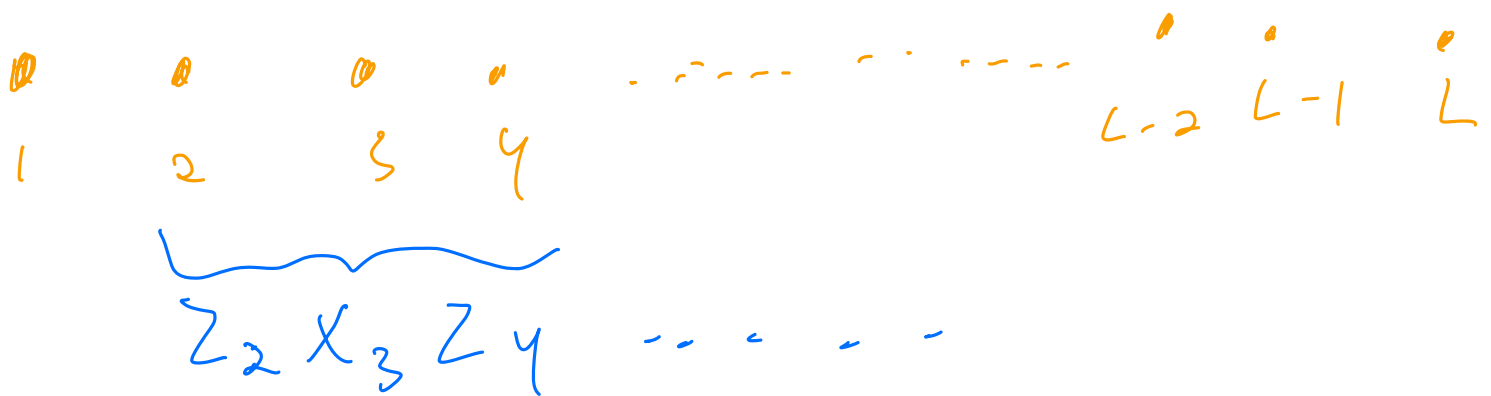
labels the projective reps of G .

Application : 1D cluster

Chain of spin $1/2$ qubits $\leftarrow 2 \text{ states}$

$$H = - \sum_{i=2}^{L-1} Z_{i-1} X_i Z_{i+1}$$

$\uparrow$ terms related to ends are missing



This chain has a global $\mathbb{Z}_2$ Symmetry

$$G = \prod_{i=1}^L X_i \quad \mathbb{Z}_2 = \{1, g\} \quad g^2 = 1$$

However cohomology is trivial $H^2(\mathbb{Z}_2, U(1)) = 0$
 it can transform it to a product of pins $H = \prod X_i$

$$W(g, g) = e^{i\theta} \quad \text{so } U(g) \rightarrow e^{i\theta/2} U(g)$$

$\mathbb{Z}_2 \times \mathbb{Z}_2 \text{ Sym}^0$	$G_{\text{even}} = \prod_{i \text{ even}} X_i$
$[H, G_{e/o}] = 0$	$G_{\text{odd}} = \prod_{i \text{ odd}} X_i$
$G_{\text{even}}^2 = G_{\text{odd}}^2 = 1$	$G_e G_o = G_o G_e$

$$H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U(1)) = \mathbb{Z}_2$$

Phases appear at the boundary

$$G_{\text{even}} \sim U_L(a) U_R(a)$$

$$G_{\text{odd}} \sim U_L(b) U_R(b)$$

$$U_L(a) \approx X, \mathbb{Z}_2 \quad U_L(b) \approx Z, \mathbb{Z}_1$$

$$U_L(g) U_L(h) = \omega(g, h) U_L(g, h)$$